

Lab-1

State Space and Combinatorial Explosion

Imagine two children playing with two water jugs. One child has a big jug that can hold 4 litres of water, called Jug X, and the other child has a small jug that can hold 3 litres of water, called Jug Y. At the beginning of the game, both jugs are completely empty because no water has been poured into them yet. Therefore, the first state is written as $(0, 0)$. The first number always represents the amount of water in Jug X, and the second number represents the amount of water in Jug Y.

The children are given a challenge. They must get exactly 2 litres of water in Jug X. They are not allowed to use a measuring cup, or any other tool to measure the water. They cannot estimate or guess the amount either. They can only solve this puzzle using the six given rules. These six rules are the only actions they are permitted to perform throughout the game.

The first rule is to fill Jug X completely. Since Jug X can hold only 4 litres, it becomes full with exactly 4 litres of water. The second rule is to fill Jug Y completely, making it hold exactly 3 litres. The third rule is to empty Jug X, removing all water from it. The fourth rule is to empty Jug Y, making it completely again. The fifth rule is to pour water from Jug X into Jug Y. The children continue pouring until Jug Y becomes full or Jug X becomes empty. They cannot stop pouring in the middle. Similarly, the sixth

rule is to pour water from Jug Y into Jug X, continuing until Jug X becomes full or Jug Y becomes empty.

The game begins at the state $(0,0)$. The children first think about everything they can do from this position. They try every one of the six rules. Some rules do not change anything because both jugs are empty. For example, emptying an already empty jug or trying to pour water from an empty jug will not create a new state. However, filling Jug X creates the state $(4,0)$ and filling Jug Y creates the state $(0,3)$. These become the first new states discovered by the children.

Every different combination of water in the two jugs is called a state. A state simply tells us how much water is present in each jug at a particular moment. For example, $(4,3)$ means both jugs are completely full. $(1,3)$ means Jug X has 1 litre while Jug Y has 3 litres. Every different combination represents a different state in the puzzle.

After discovering the first few states, we continue exploring. From $(4,0)$ they again perform all six operations. Filling Jug X again does not create anything new because it is already full. Emptying Jug X brings them back to $(0,0)$. Filling Jug Y creates $(4,3)$. Pouring water from Jug X into Jug Y creates $(1,3)$ because Jug Y can only hold 3 litres. Every operation is checked carefully to see whether it produces a new state.

The same process is repeated for $(0,3)$. From there they can empty Jug Y, fill Jug X, pour water into Jug X, or perform the remaining operations. Again, every possible move is explored. Instead of trying only one path, they try all possible paths. This ensures that no possible solution is missed.

Now imagine one of the children is carrying a notebook. Whenever a new state is discovered, it is written in the notebook. Before writing any state, the child first checks whether it has already been written. If it's already there, it is not written again. This notebook is very important because it helps the children remember every state already visited.

For example, suppose the children are at $(4,0)$. If they empty Jug X, they return to $(0,0)$. Since $(0,0)$ is already written in the notebook, they do not waste time exploring it again. In the same way, many operations lead back to states that have already been visited. Because the notebook remembers every explored state, the children avoid repeating the same work again and again.

Another interesting part of the game is that the children do not explore randomly. They explore level by level. At the beginning, there is only one state $(0,0)$. From this, they discover a few new states. These new states form the first level. Then they explore every state in the first level before moving to the second level. After finishing the

second level, they move to the third level, and so on. This organized method ensures that all nearby possibilities are explored before moving farther away.

Imagine, a tree with branches. If every state always produced six completely new states, then after one level there would be six states. After the third level there would be 216 states, and after the fourth level, there would be 1296 states. The numbers become larger and larger very quickly. This rapid increase is called combinatorial explosion because the number of paths grows like an explosion.

However, this does not happen in water jug problem. Many of the branches lead back to state that have already been explored. Instead of creating thousands of completely different states the program recognizes repeated states and ignores them. This keeps the search small and efficient.